NSCAS NEBRASKA STUDENT-CENTERED ASSESSMENT SYSTEM

Grade 5 General Science

Directions:

On the following pages of your practice test booklet are questions for the Grade 5 *Nebraska Student-Centered Assessment System (NSCAS) General Science* practice test.

Read these directions carefully before beginning the test.

This practice test will include several different types of questions. Multiple choice questions will ask you to select an answer from among several answer choices. Multiple select questions will ask you to select multiple correct answers from among five or more answer choices. For some questions, there may be two parts, Part A and Part B. Some questions will ask you to construct an answer by following the directions given.

For all questions:

- Read each question carefully and choose the best answer.
- Review each question carefully. You may not return to a previous question.
- You may use scratch paper to make notes.
- Be sure to answer ALL the questions.

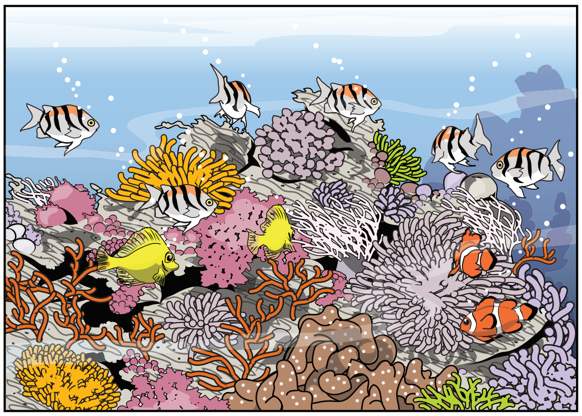
STOP.

1. **Use the information to answer the question.**

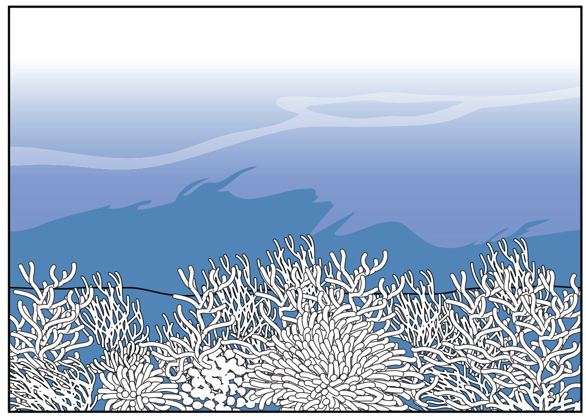
Changing Coral

Scientists have been investigating coral reef ecosystems over time. They have observed many changes.

Coral Reef before 1980



Coral Reef Today



In some areas, the bright colors of the coral reef have disappeared. This is called **coral bleaching**. The coral's colors come from plant-like algae living in the coral. Students want to know more about how coral bleaching changes the relationships between organisms in the coral reef.

In which resource might the students find information about what is causing the changes to the coral reef?

A. historical maps of the coral reef

B. pictures of the coral reef from a family vacation

C. weather forecast reports for the coral reef today

D. scientific journal articles on the marine biology of coral reefs

Review each question carefully. You may not return to a previous question.

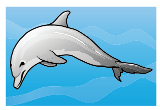
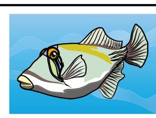
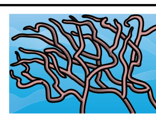
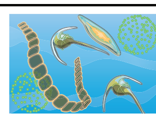
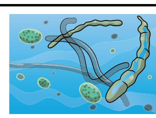
2. **Use the information to answer the question.**

Changing Coral

Students research corals and find out that corals and zooxanthellae, a microscopic algae, work together to survive. Zooxanthellae make nutrients through photosynthesis that feed the corals. The corals then give shelter to the algae.

Students collect information from different sources and create a list of the organisms in the coral reef.

Reef Organisms

Organism	Picture	How It Obtains Energy
dolphin		eats fish
triggerfish		eats coral
coral		from algae
algae		from the sun
bacteria		from dead organisms

A student hypothesizes that consumers are most affected by coral bleaching.

Which organism in the table is a producer?

A. coral

B. algae

C. dolphin

D. triggerfish

Review each question carefully. You may not return to a previous question.

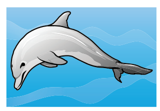
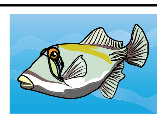
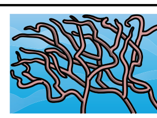
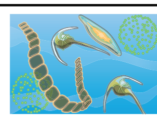
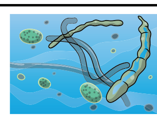
3. **Use the information to answer the question.**

Changing Coral

Students research corals and find out that corals and zooxanthellae, a microscopic algae, work together to survive. Zooxanthellae make nutrients through photosynthesis that feed the corals. The corals then give shelter to the algae.

Students collect information from different sources and create a list of the organisms in the coral reef.

Reef Organisms

Organism	Picture	How It Obtains Energy
dolphin		eats fish
triggerfish		eats coral
coral		from algae
algae		from the sun
bacteria		from dead organisms

A student hypothesizes that consumers are most affected by coral bleaching.

Go on to the next page.

This question has two parts. Answer Part A, and then answer Part B.

Part A

Which type of model should the student make to show the relationships between the organisms?

A. a T-chart

B. a food web

C. a bar graph

D. a circle chart

Part B

Which parts of the system should be included in the model chosen in the answer to Part A?

Select **four** parts.

A. producers

B. consumers

C. decomposers

D. carbon dioxide

E. population size

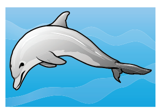
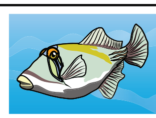
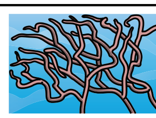
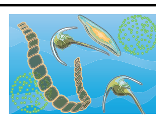
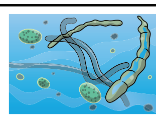
F. energy transferred

4. **Use the information to complete the task.**

Changing Coral

Scientists have been investigating coral reef ecosystems over time. They have observed many changes. In some areas, the bright colors of the coral reef have disappeared. The coral's colors come from plant-like algae living in the coral. Students want to know more about how coral bleaching changes the relationships between organisms in the coral reef.

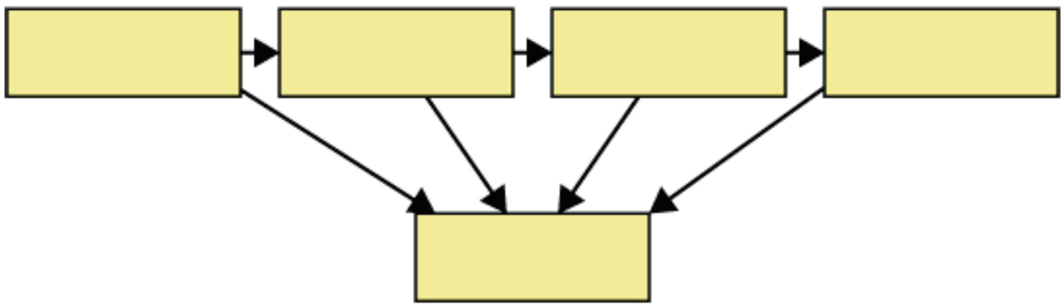
Reef Organisms

Organism	Picture	How It Obtains Energy
dolphin		eats fish
triggerfish		eats coral
coral		from algae
algae		from the sun
bacteria		from dead organisms

Complete the model of the ocean ecosystem to describe how matter moves among the five organisms.

Write each organism in a box in the model.

dolphin triggerfish coral algae bacteria

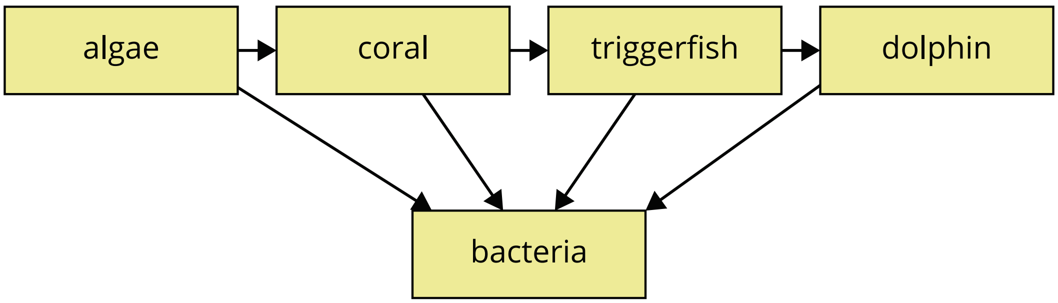


Review each question carefully. You may not return to a previous question.

5. **Use the information to answer the question.**

Changing Coral

Scientists make this model of a coral ecosystem to predict how coral bleaching will affect the organisms.



How will the populations of the organisms be affected first if the population of coral algae greatly decreases?

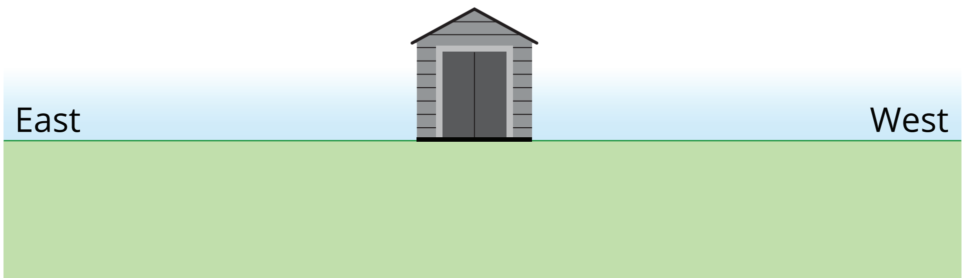
Select the effect on each organism's population.

Organism	Population Effect		
dolphin	increase	decrease	little to no effect
triggerfish	increase	decrease	little to no effect
coral	increase	decrease	little to no effect

6. **Use the information to answer the question.**

Garden Shadows

Students are going to build a community garden next to the garden shed at their school. The students want to know the best place to put the garden beds so they receive afternoon sun. The picture shows the shed.



Where is the sun located in the morning in relation to the shed?

A. to the east

B. to the west

C. to the north

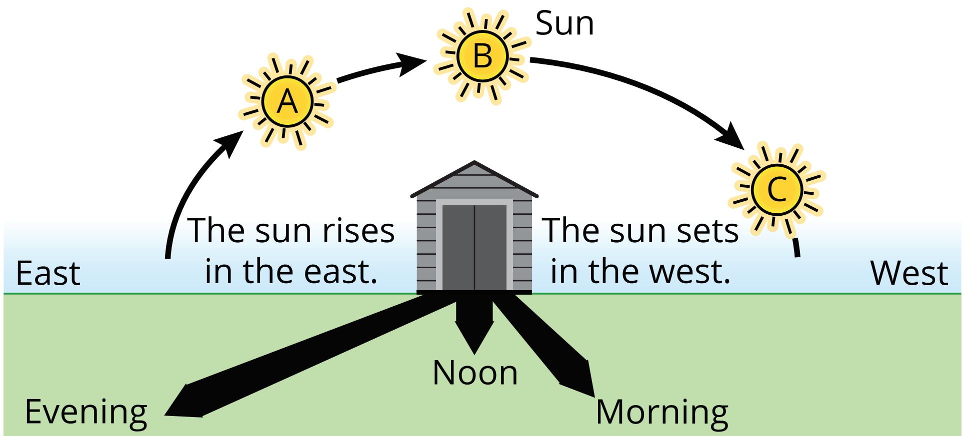
D. to the south

7. **Use the information to complete the task.**

Garden Shadows

A group of students wants to test the way the shadows affect the different locations around the shed. They take measurements of shadows around the shed throughout the day and draw a model.

Shadows of a Shed



Explain how shadows move throughout the day.

Select the correct choices to complete the sentences.

When the sun is rising, the shadow is pointing (**east**, **west**).

When the sun is setting, the shadow is pointing (**east**, **west**).

During the day, the shadow moves (**east**, **west**) to (**east**, **west**).

8. **Use the information to answer the question.**

Garden Shadows

The students measure the length of the shed's shadows throughout a day. They record their data in a table.

Length of a Shadow During the Day in May

Time of Day	**Length of Shed's Shadow** (meters)
7:30 a.m.	21.4
9:30 a.m.	9.5
11:30 a.m.	5.6
1:00 p.m. (The sun is at its highest point in the sky at 1:20 p.m.)	4.5
3:00 p.m.	5.2
4:00 p.m.	6.5
6:00 p.m.	11.6
8:00 p.m. (Sunset is at 8:35 p.m.)	31.9

Complete the sentences to provide evidence that supports how shadow length shows patterns during the day.

Write the correct time of day in each blank. Options may be used more than once.

7:30 a.m. 1:20 p.m. 8:35 p.m.

Between 8:00 a.m. and ____________, the length of the shadow decreases.

Then, between ____________ and ____________, the length of the shadow increases.

Review each question carefully. You may not return to a previous question.

9. **Use the information to answer the question.**

Garden Shadows

The students measure the length of the shed's shadows throughout a day. They record their data in a table.

Length of a Shadow During the Day in May

Time of Day	**Length of Shed's Shadow** (meters)
7:30 a.m.	21.4
9:30 a.m.	9.5
11:30 a.m.	5.6
1:00 p.m. (The sun is at its highest point in the sky at 1:20 p.m.)	4.5
3:00 p.m.	5.2
4:00 p.m.	6.5
6:00 p.m.	11.6
8:00 p.m. (Sunset is at 8:35 p.m.)	31.9

Which statement BEST describes a pattern in shadow length throughout a day?

A. A shadow is shortest in the early morning.

B. A shadow is longest in the late afternoon.

C. A shadow is shortest when the sun is closest to the horizon and longest when the sun is farthest from the horizon.

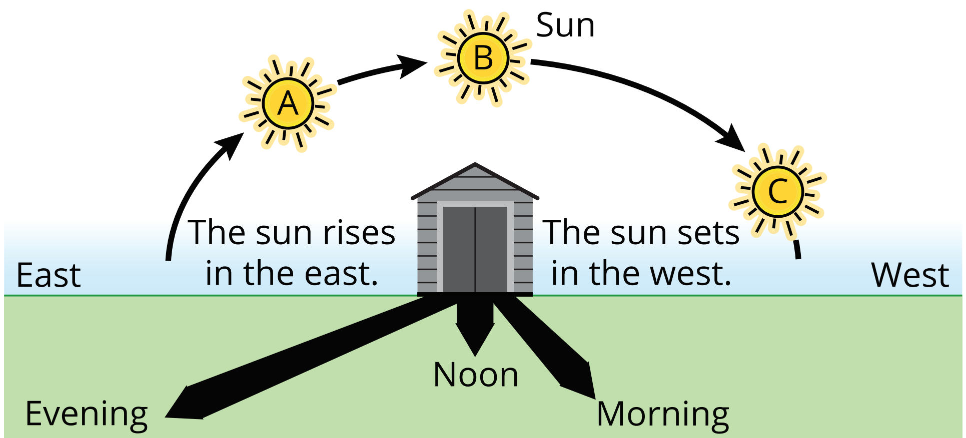
D. A shadow is longest when the sun is closest to the horizon and shortest when the sun is farthest from the horizon.

10. **Use the information to complete the task.**

Garden Shadows

Students are going to build a community garden next to the garden shed at their school. The students want to know the best place to put the garden beds so they receive afternoon sun. The picture shows the shed.

Shadows of a Shed



The students notice that the shed's shadow falls to the west in the morning.

This question has two parts. Answer Part A, and then answer Part B.

Part A

What data will the students use to decide where to plant the garden?

A. height of shed

B. hours of daylight

C. location of shadows

D. average daily temperature

Part B

Where should the garden beds be planted to get the most afternoon sun?

A. to the east

B. to the west

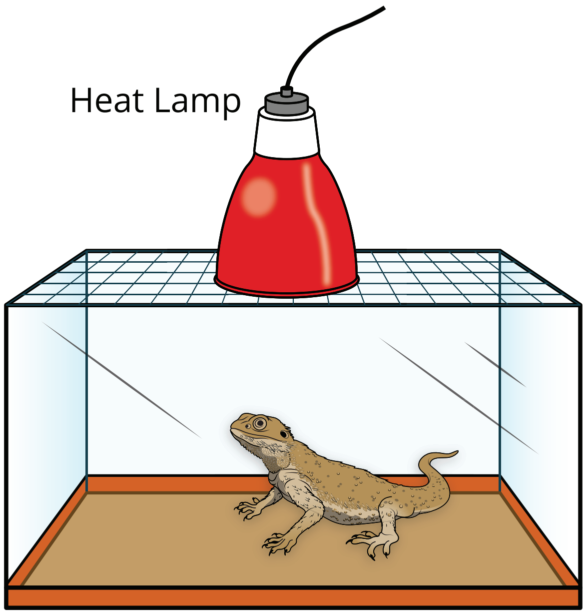
C. to the north

D. to the south

11. **Use the information to answer the question.**

Designing a Lizard Habitat

A science class has a lizard as a class pet. The lizard is not staying warm enough in its habitat.



The students want to design a better habitat to maintain the proper temperature for the lizard. They want to change the material at the bottom of the habitat to collect more heat from the heat lamp.

Which investigation would help the students identify the best material for the bottom of the habitat?

A. Observing the texture of different materials.

B. Measuring the height of the different walls of the habitat.

C. Recording the temperature of the cage in four different locations.

D. Comparing how different materials affect the temperature of the habitat.

12. **Use the information to answer the question.**

Designing a Lizard Habitat

Students decide to compare how different materials placed on the bottom of the tank affect the temperature of the lizard habitat.

Materials can be identified by their properties. One property is how materials conduct heat. To compare the different materials, the students conduct an investigation to measure how quickly heat flows through four different materials that could be used in the lizard habitat.

Which tool should be used to measure how long it takes for heat to flow through the different materials?

A. a ruler

B. a balance

C. a stopwatch

D. a microscope

13. **Use the information to answer the question.**

Designing a Lizard Habitat

Materials can be identified by their properties. One property is how materials conduct heat. To compare the different materials, the students conduct an investigation to measure how quickly heat flows through four different materials that could be used in the lizard habitat.

Students use four containers made of four different types of materials. They place one ice cube in each container and measure the amount of time it takes to melt. For the four materials:

- Each container is the same volume.
- The ice cubes are the same size.
- Observations are collected every minute.
- The total time it takes for the ice to melt is recorded.

Go on to the next page.

This question has two parts. Answer Part A, and then answer Part B.

Part A

Which variables in the experiment are controlled?

Select **three** variables.

A. size of ice cube

B. size of container

C. number of ice cubes

D. type of container material

E. time it takes the ice to melt

Part B

Which variable is tested in the investigation?

A. how the size of the ice cube affects how quickly it melts

B. how the number of ice cubes changes based on container size

C. how long it takes ice to melt in containers made of different materials

D. how the room temperature changes when ice is added to the containers

14. **Use the information to answer the question.**

Designing a Lizard Habitat

The students record their data from their lizard habitat investigation. Table 1 describes the observable properties of the four different containers used in the investigation that were made of materials A, B, C, and D. It shows how long it takes one ice cube to melt in each container.

Table 1. Investigation Data

Container	**Property of Material**	**Time to Melt** (minutes)
A	• clear, blue color • stiff	6
B	• white color • bends slightly • goes back into shape without breaking	9
C	• light brown • unbendable • bending results in a crack	4
D	• gray • thin, bendable • tears	7

Conductivity is how easily heat can move through a material. Materials with high conductivity will lose heat much faster than materials with low conductivity.

Based on Table 1, which material has the LOWEST conductivity?

A. material A

B. material B

C. material C

D. material D

15. **Use the information to answer the question.**

Designing a Lizard Habitat

The students identify container B as the material with the lowest conductivity. They want to use this material for the lizard's habitat so that it stays warmer.

Table 1. Properties of Drinking Containers

Container	**Property of Material**	**Time to Melt** (minutes)
B	• white color • bends slightly • goes back into shape without breaking	9

Table 2 shows the properties of three different materials that can be used to make the bottom of lizard habitats.

Table 2. Properties of Lizard Habitat Materials

Material	**Properties**
Foam	• extremely light weight and flexible • usually white • very low heat flow
Newspaper	• light weight material • usually gray • medium heat flow
Plastic	• light weight material • may be transparent • medium heat flow
Rubber	• medium weight material • dark in color • high heat flow

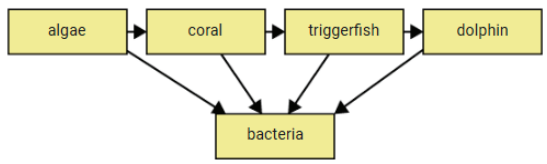
Select the correct words to complete the sentence(s).

Container B is made of (**foam**, **newspaper**, **plastic**, **rubber**).

The bottom of the lizard's habitat should be made from this material because it is the (**highest**, **lowest**) conductor of heat energy from the heat lamp.

STOP.

NSCAS General Grade 5
Practice Test Answer Key
Science (English)

Sequence	Key	Points
1	**D**	1
2	**B**	1
3	Part A: **B** Part B: **A, B, C, F**	2
4		1
5	**decrease, decrease, decrease**	1
6	**A**	1
7	**west, east, west, east**	1
8	**1:20 p.m., 1:20 p.m., 8:35 p.m.**	1
9	**D**	1
10	Part A: **C** Part B: **B**	2
11	**D**	1
12	**C**	1
13	Part A: **A, B, C** Part B: **C**	2
14	**B**	1
15	**foam, lowest**	2